会算术的马

The Horse That Could Count



汉斯是在算数，还是在读懂人的动作?

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

Wilhelm von Osten, a mathematics teacher and trainer in Berlin, owned a stallion, Hans. Asked to calculate a sum, attempt subtraction, or name a date, Hans tapped out the answer with a hoof. Such aptitude suggested brilliance, and demonstrations in his owner's courtyard made him seem intelligent. The New York Times reported the stunning act in 1904.

A commission of thirteen was formed to investigate. It included a veterinarian, a circus manager, a cavalry officer, schoolteachers, and the director of the Berlin zoological garden. Its members looked for trickery and collusion. In September 1904 they found no deception, in effect certifying the act authentic.

In 1907 the psychologist Oskar Pfungst tested the act systematically, altering one variable after another. He kept onlookers away, rotated questioners, fitted blinders, and sometimes used people ignorant of the answer. When Hans could see the questioner, he answered correctly 50 times in 56. With the questioner invisible, his accuracy fell to two in 35.

The cues came from the questioners. Each would lean forward as tapping began, and tension rose as the taps neared the answer. At the right tap it eased, and the questioner would straighten or jerk the head upward. Every gesture and motion was subconscious, made unwittingly by people with no wish to deceive.

Double-blind designs in psychology and animal cognition guard against the Clever Hans effect, a phenomenon of bias in which expectancy shapes behavioural results. In the laboratory, Pfungst took the animal's place and proved sensitive to the cues of attentive volunteers. Hans went, on his owner's death in June 1909, to a jeweller still convinced that such animals reasoned.

中文译文

柏林的数学教师兼驯马师威廉·冯·奥斯滕，养着一匹名叫汉斯的公马。让它计算加法、做减法，或者说出某个日期时，汉斯就会用一只蹄子敲出答案。这种本领看起来显示了非凡的才智；它在主人庭院里的表演，也让人觉得它很聪明。1904年，《纽约时报》报道了这场令人惊叹的表演。

一个由13人组成的委员会随后成立，负责调查。成员包括兽医、马戏团经理、骑兵军官、中小学教师，以及柏林动物园园长。委员们查找是否存在欺骗手法或暗中串通。1904年9月，他们没有发现有意欺骗的证据，实际上等于认可了表演的真实性。

1907年，心理学家奥斯卡·普丰斯特系统地检验了这项表演，逐一改变实验变量。他让围观者远离现场，轮换提问者，给马戴上遮眼罩，有时还请不知道答案的人提问。汉斯能够看见提问者时，56次作答中有50次正确；看不见提问者时，35次里却只对了2次。

提示来自提问的人。马蹄开始敲击时，人会向前倾身；次数越接近正确答案，人就越紧张。敲到正确的那一下时，紧张感便缓和下来，提问者会挺直身子，或者猛地抬一下头。这些姿态和动作全都是下意识的，是无意中做出来的；提问者并没有欺骗别人的打算。

在心理学和动物认知研究中，双盲实验设计可用来防范“聪明汉斯效应”：研究者的预期通过无意暗示，影响了所观察到的行为结果。在实验室里，普丰斯特还亲自代替那匹马，证明自己也能敏锐地察觉专注的志愿者发出的细微信号。汉斯的主人于1909年6月去世后，它归了一位珠宝商所有；那位珠宝商依然相信，这类动物能够进行推理。

没有发现有意作弊，并不等于已经证明马能够做算术。

核心词汇

先记这22项。连着下方的原文短语一起回想。

aptitude n.
天资；才能
Such aptitude suggested brilliance,

commission n.
调查委员会
A commission of thirteen

investigate v.
调查
to investigate.

collusion n.
暗中串通
trickery and collusion

deception n.
欺骗
they found no deception

certify v.
确认；证明
certifying the act authentic

authentic adj.
真实可信的
certifying the act authentic

systematically adv.
系统地；有步骤地
tested the act systematically

variable n.
变量
altering one variable after another.

accuracy n.
准确率；准确性
his accuracy fell to two in 35.

cue n.
提示；信号
The cues came from the questioners.

tension n.
紧张
and tension rose

gesture n.
手势；姿态
Every gesture and motion

subconscious adj.
下意识的
motion was subconscious

unwittingly adv.
无意中；不知不觉地
made unwittingly by people

deceive v.
欺骗
no wish to deceive

double-blind adj.
双盲的
Double-blind designs

cognition n.
认知
animal cognition

phenomenon n.
现象
a phenomenon of bias

bias n.
偏差；偏倚
a phenomenon of bias

expectancy n.
预期；期待心理
expectancy shapes behavioural results.

sensitive adj.
敏感的；能察觉细微变化的
proved sensitive to the cues

文中词汇与搭配

补充词汇共28项。保留每个表达在本文中的意思。

trainer n.
驯马师；训练者

stallion n.
公马

calculate v.
计算

subtraction n.
减法

brilliance n.
非凡才智

demonstration n.
演示；表演

courtyard n.
庭院

intelligent adj.
聪明的

stunning adj.
令人惊叹的

veterinarian n.
兽医

cavalry n.
骑兵

zoological adj.
动物学的

trickery n.
欺骗手法

psychologist n.
心理学家

rotate v.
轮换

questioner n.
提问者

blinder n.
马用遮眼罩

ignorant adj.
不知道的

correctly adv.
正确地

invisible adj.
不可见的；看不见的

lean v.
倾身；倾斜

straighten v.
挺直身子

jerk v.
猛地一抬；急动

motion n.
动作

psychology n.
心理学

behavioural adj.
行为的

attentive adj.
专注的

jeweller n.
珠宝商

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In 1907 the psychologist Oskar Pfungst tested the act systematically, altering one _____ after another.

02 With the questioner invisible, his _____ fell to two in 35.

03 The _____ came from the questioners.

04 Every gesture and motion was _____, made unwittingly by people with no wish to deceive.

文章主线

- 奇观 柏林一匹名叫汉斯的马用蹄子敲出算式答案；早期调查没发现作弊，表演因而被视为可信。
- 检验 心理学家普丰斯特改变视线等条件：能看见提问者时56次对50次，看不见时35次仅对2次。
- 机制 提问者会无意中在答案处抬头或放松；汉斯识别这些身体信号来停止敲蹄，被误以为会算术。
- 复现 普丰斯特亲自代替汉斯，也能根据志愿者的细微动作作答，说明这种影响不限于马。
- 防范 研究者的期待可经无意暗示影响结果，这叫聪明汉斯效应；双盲等实验设计用来减少这种偏差。

参考答案　1 variable 2 accuracy 3 cues 4 subconscious